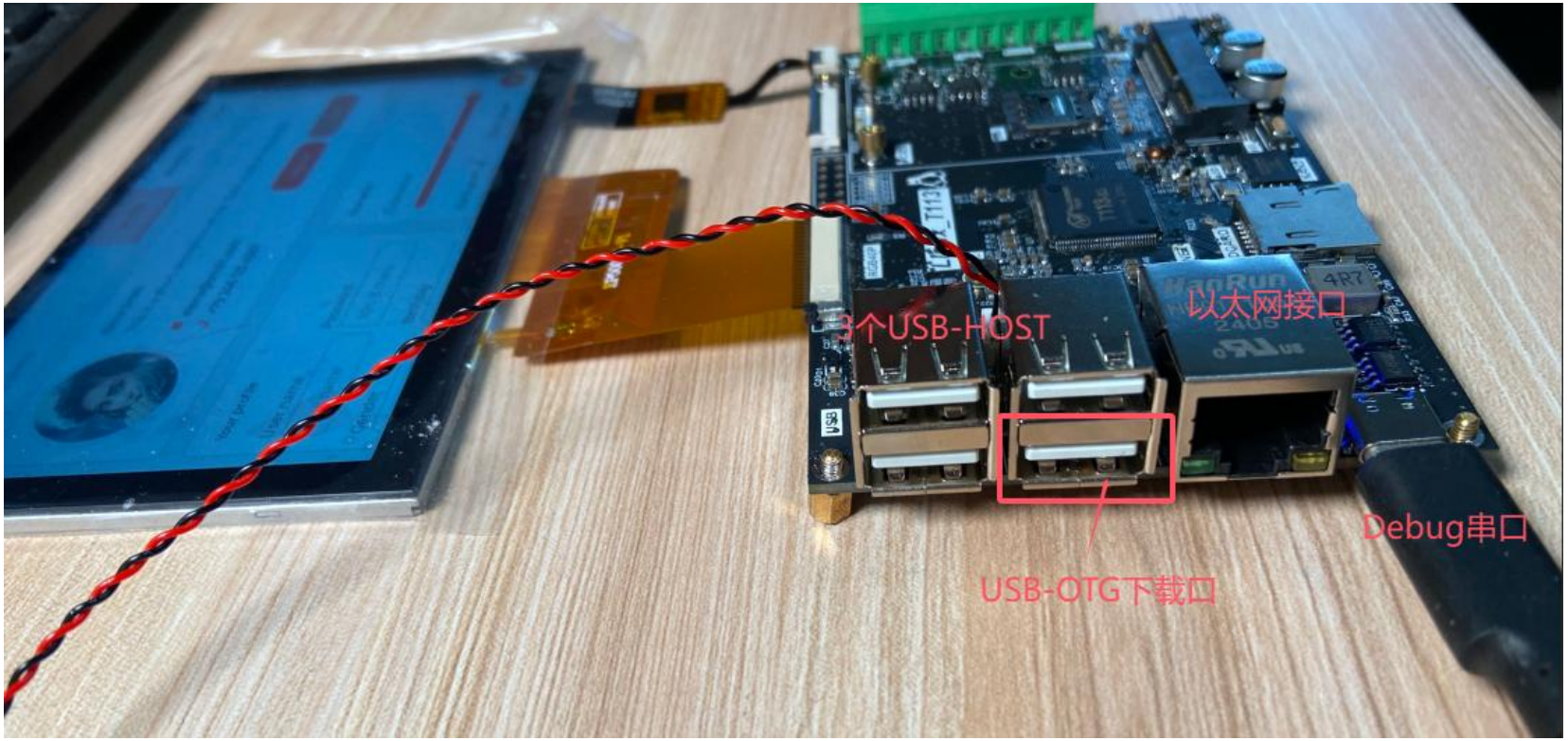
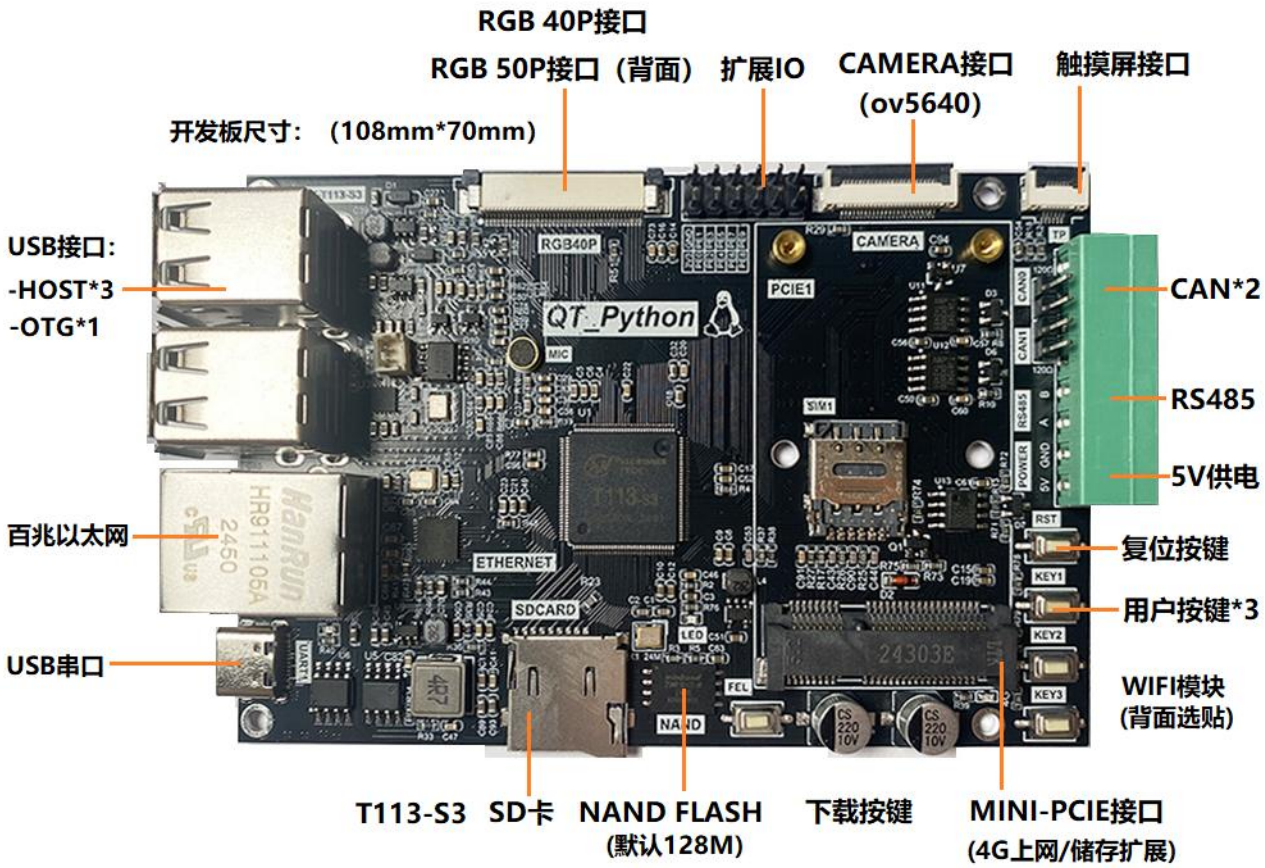
T113 开发文档 - 开箱上手篇

一. 开发板接口说明：



开发板尺寸及接口


RGB 40P接口
RGB 50P接口（背面）
扩展IO
CAMERA接口
(ov5640)
触摸屏接口
开发板尺寸：（108mm*70mm）
USB接口：
-HOST*3
-OTG*1
CAN*2
RS485
5V供电
百兆以太网
复位按键
用户按键*3
USB串口
WIFI模块
(背面选贴)
T113-S3
SD卡
NAND FLASH
(默认128M)
下载按键
MINI-PCIE接口
(4G上网/储存扩展)

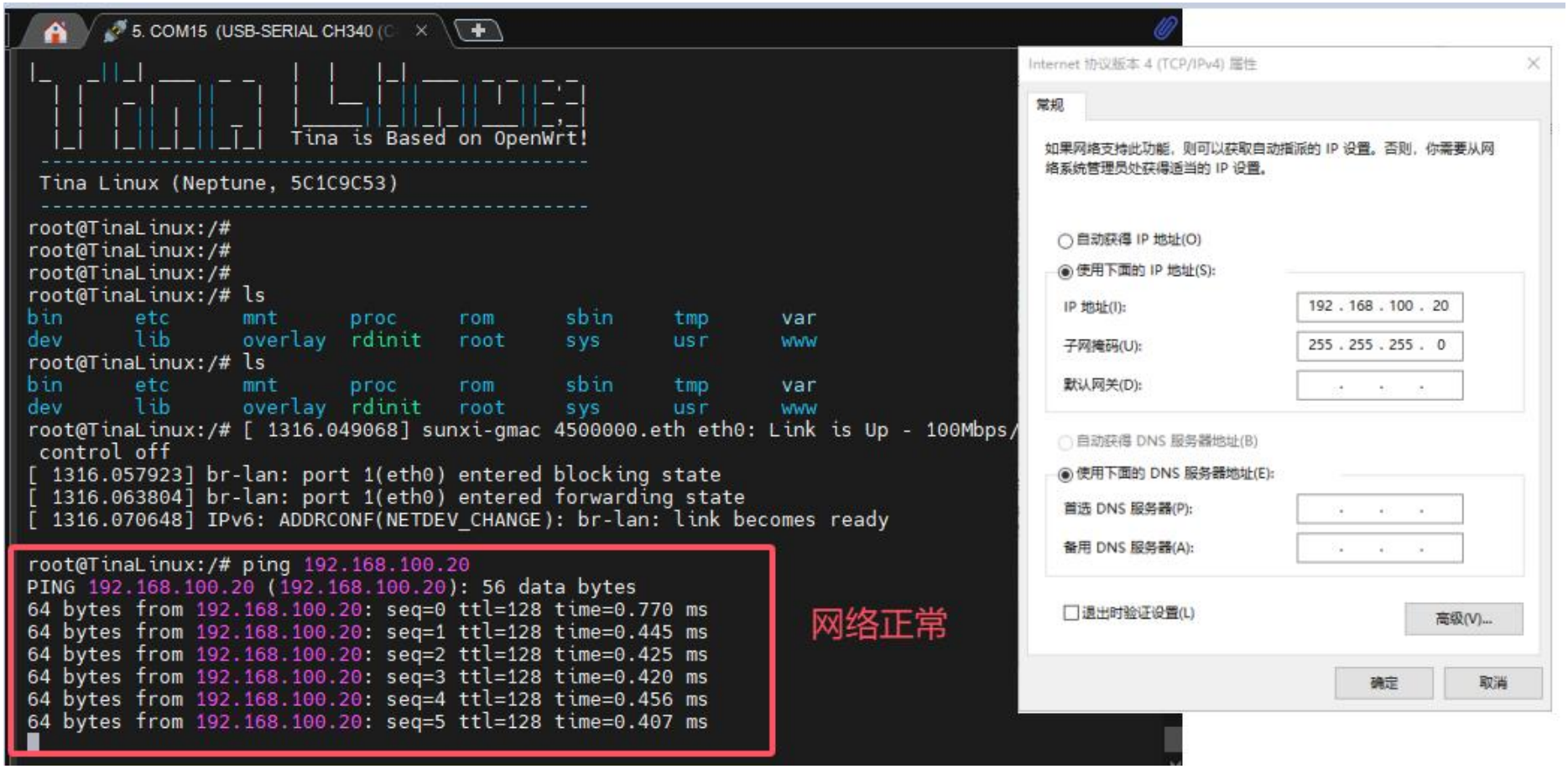
二. 软件测试

1. 串口测试：（接 Type-C 接口，波特率 115200）

```
5. COM15 (USB-SERIAL CH340 (C
[   20.925627] RTW: cfg80211_rtw_get_txpower(wlan0) total max: -10000 mbm
[   20.933102] RTW: rtw_ndev_notifier_call(wlan0) state:4
[   20.989521] RTW: cfg80211_rtw_flush_pmksa(wlan0)
[   33.808889] usb1-vbus: disabling
[  164.268879] sunxi_vbus_det_work()3454 WARN: get power supply failed
[  164.276231] android_work: sent uevent USB_STATE=DISCONNECTED


BusyBox v1.27.2 () built-in shell (ash)

 _____ _                 __      _
|_   _||_| ___  _ _    |  |    |_| ___  _ _  _ _
  | |    _ |   ||  |   |  |__ | ||   || | ||_'_|
  | |   | || | || _ |  |_____||_||_|_||___||_,_|
  |_|   |_||_|_||_|_|  Tina is Based on OpenWrt!
 ----------------------------------------------
 Tina Linux (Neptune, 5C1C9C53)
 ----------------------------------------------
root@TinaLinux:/#
root@TinaLinux:/#
root@TinaLinux:/#
root@TinaLinux:/# ls
bin      etc      mnt      proc     rom      sbin     tmp      var
dev      lib      overlay  rdinit   root     sys      usr      www
root@TinaLinux:/# ls
bin      etc      mnt      proc     rom      sbin     tmp      var
dev      lib      overlay  rdinit   root     sys      usr      www
root@TinaLinux:/#
```

2. 以太网测试

板子 IP 是 192.168.100.100，需要将 PC 设置成 192.168.100.20），串口输入 ping 192.168.100.20 测试网络是否 OK



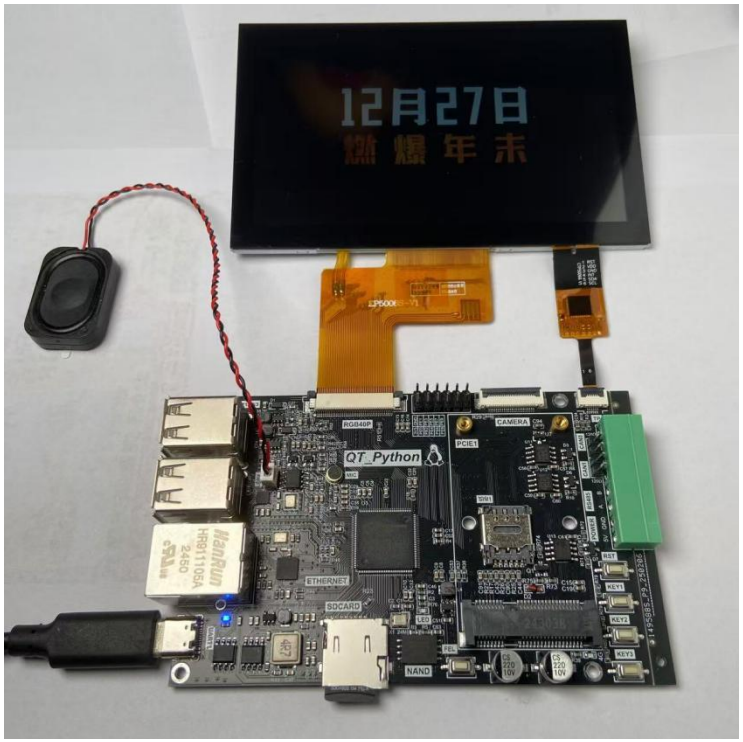
3. 显示屏测试

软件已预装 lvgl demo, 接上屏幕（断电操作），串口执行指令“lv_examples 0”即可显示（预装系统已设置自动启动）；

播放视频测试：放入视频文件到 U 盘根目录 xxx.mp4

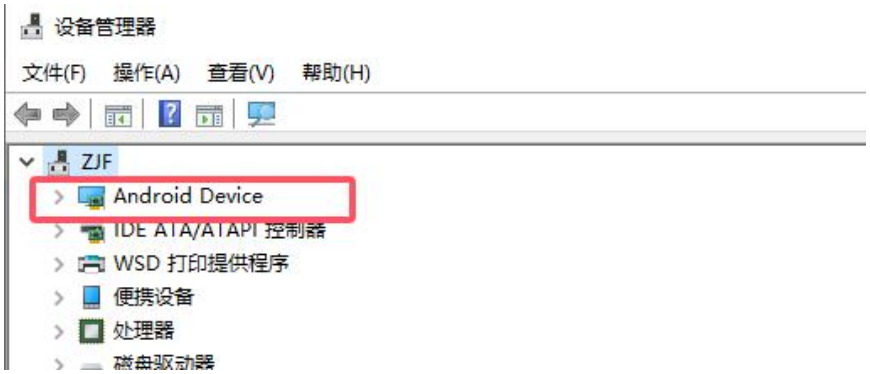
播放视频

```
tplayerdemo /mnt/exUDISK/xxx.mp4
```



4. ADB 测试

开机后接入 adb 线到 OTG 口，设备管理器会识别到 adb 设备，需要安装 Google adb 驱动；



输入 adb shell 可以像串口一样操作，主要用来 push/pull 文件，需要配置下 adb 环境变量

```
C:\Users\Administrator>adb shell
* daemon not running. starting it now on port 5037 *
* daemon started successfully *

BusyBox v1.27.2 () built-in shell (ash)

 Tina is Based on OpenWrt!
 -----------------------------------------------
 Tina Linux (Neptune, 5C1C9C53)
 -----------------------------------------------
root@TinaLinux:/# ls
ls
bin      etc      mnt      proc     rom      sbin     tmp      var
dev      lib      overlay  rdinit   root     sys      usr      www
root@TinaLinux:/#
```

5. WIFI 连接（需要选择 wifi 版本硬件）

5.1 修改 wifi 配置文件 ssid 和 psk：/etc/wpa_supplicant.conf

文件内容如下：

```
# 修改 ssid 为你的路由器名，psk 为密码
ctrl_interface=/var/run/wpa_supplicant

network={
        ssid="TP-LINK_123"
        psk="12345678"
        key_mgmt=WPA-PSK
        proto=WPA2
        pairwise=CCMP
        group=CCMP
}
```

6.2 连接 WIFI ： wpa_supplicant -i wlan0 -Dnl80211 -c /etc/wpa_supplicant.conf -B

6.3 自动分配 ip 地址：udhcpc -i wlan0 &

6.4 ping 百度测试： ping www.baidu.com

注意：WIFI 模块和 4G 模块共用了同一个 USB，测试 WIFI 时不要接 4G 模块；

```
BusyBox v1.27.2 () built-in shell (ash)

 _____ _____ _____ _____ _____ _____ _____ _____
|_   _|_   _|_   _|_   _|_   _|_   _|_   _|_   _|
  | |   | |   | |   | |   | |   | |   | |   | |
  |_|   |_|   |_|   |_|   |_|   |_|   |_|   |_|
 ---------------------------------------------
 Tina Linux (Neptune, 5C1C9C53)
 ---------------------------------------------
root@TinaLinux:/#

root@TinaLinux:/#

root@TinaLinux:/# wpa_supplicant -i wlan0 -Dnl80211 -c /etc/wpa_supplicant.conf -B
wpa_supplicant -i wlan0 -Dnl80211 -c /etc/wpa_supplicant.conf
-B
Successfully initialized wpa_supplicant
root@TinaLinux:/# udhcpc -i wlan0 &
udhcpc -i wlan0 &
root@TinaLinux:/# udhcpc: started, v1.27.2
udhcpc: sending discover
udhcpc: sending select for 192.168.3.58
udhcpc: lease of 192.168.3.58 obtained, lease time 86400
udhcpc: ifconfig wlan0 192.168.3.58 netmask 255.255.255.0 broadcast +
udhcpc: setting default routers: 192.168.3.1


[1]+  Done                       udhcpc -i wlan0
root@TinaLinux:/#

root@TinaLinux:/# ping www.baidu.com
ping www.baidu.com
PING www.baidu.com (183.2.172.185): 56 data bytes
64 bytes from 183.2.172.185: seq=0 ttl=52 time=42.794 ms
64 bytes from 183.2.172.185: seq=1 ttl=52 time=24.326 ms
64 bytes from 183.2.172.185: seq=2 ttl=52 time=22.592 ms
64 bytes from 183.2.172.185: seq=3 ttl=52 time=25.955 ms
64 bytes from 183.2.172.185: seq=4 ttl=52 time=26.227 ms
64 bytes from 183.2.172.185: seq=5 ttl=52 time=22.945 ms
64 bytes from 183.2.172.185: seq=6 ttl=52 time=25.100 ms
```

wifi连接

动态获取ip

ping外网测试

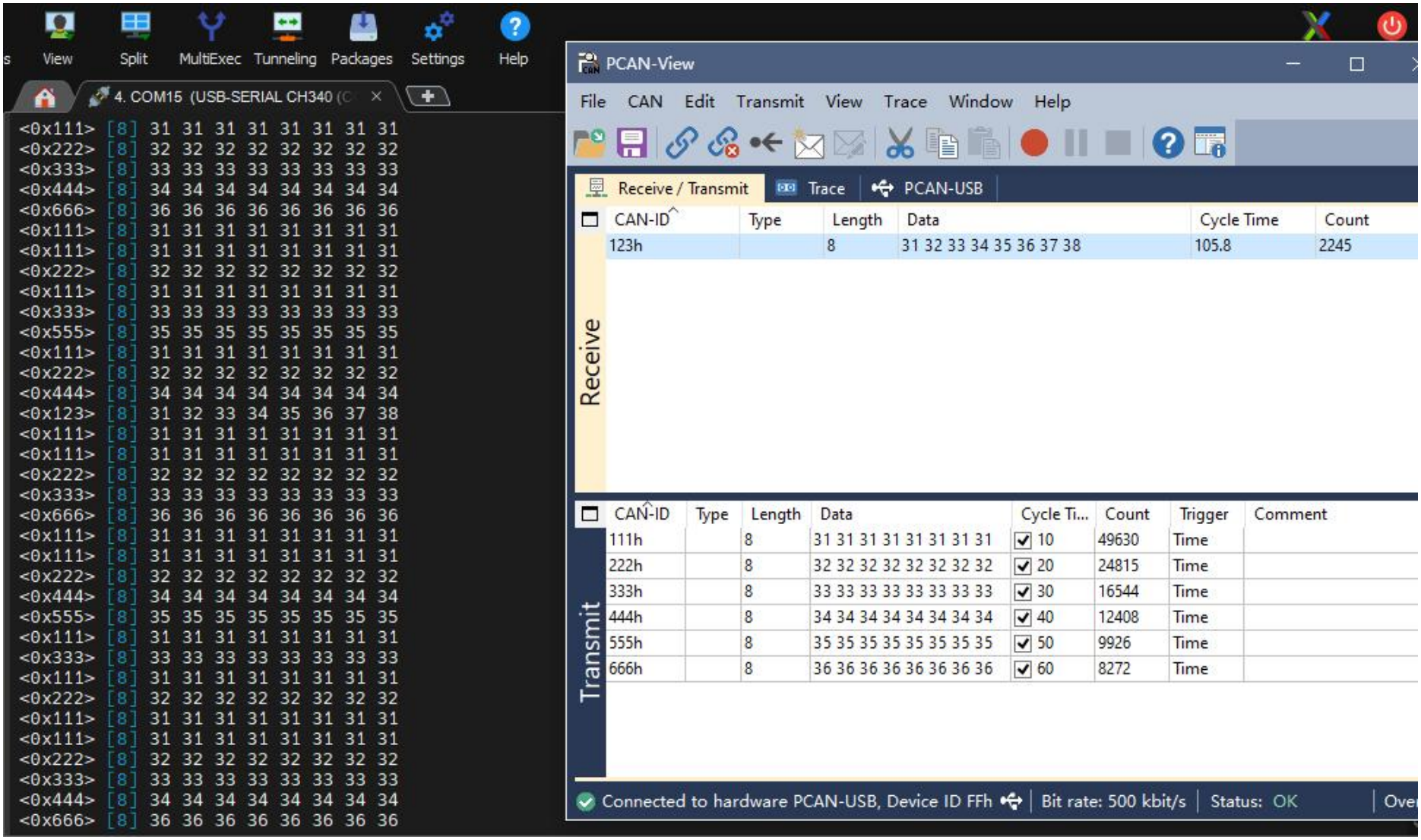
6. QT 应用测试

说明：开机默认启动的是 LVGL 应用，要启动 QT 应用，需要烧录带 qt 的镜像，或是打上附件的 patch；

```
# 先杀掉LVGL 进程:
kill -9 $(pidof lv_examples)

# 配置qt 环境:
source /etc/qt-env.sh

# 运行qt demo 程序:
qt-easing
```

7. CAN 测试：

```
### can 配置波特率并使能
ip link set can0 type can bitrate 500000
ifconfig can0 up

##发送数据
candump can0 cansend can0 -i0x123 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38

##接收数据
candump can0
```


View
Split
MultiExec
Tunneling
Packages
Settings
Help
4. COM15 (USB-SERIAL CH340
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x333> [8] 33 33 33 33 33 33 33 33
<0x444> [8] 34 34 34 34 34 34 34 34
<0x666> [8] 36 36 36 36 36 36 36 36
<0x111> [8] 31 31 31 31 31 31 31 31
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x111> [8] 31 31 31 31 31 31 31 31
<0x333> [8] 33 33 33 33 33 33 33 33
<0x555> [8] 35 35 35 35 35 35 35 35
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x444> [8] 34 34 34 34 34 34 34 34
<0x123> [8] 31 32 33 34 35 36 37 38
<0x111> [8] 31 31 31 31 31 31 31 31
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x333> [8] 33 33 33 33 33 33 33 33
<0x666> [8] 36 36 36 36 36 36 36 36
<0x111> [8] 31 31 31 31 31 31 31 31
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x444> [8] 34 34 34 34 34 34 34 34
<0x555> [8] 35 35 35 35 35 35 35 35
<0x111> [8] 31 31 31 31 31 31 31 31
<0x333> [8] 33 33 33 33 33 33 33 33
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x111> [8] 31 31 31 31 31 31 31 31
<0x111> [8] 31 31 31 31 31 31 31 31
<0x222> [8] 32 32 32 32 32 32 32 32
<0x333> [8] 33 33 33 33 33 33 33 33
<0x444> [8] 34 34 34 34 34 34 34 34
<0x666> [8] 36 36 36 36 36 36 36 36
PCAN-View
File CAN Edit Transmit View Trace Window Help
Receive / Transmit
Trace
PCAN-USB
Receive
CAN-ID Type Length Data Cycle Time Count
123h 8 31 32 33 34 35 36 37 38 105.8 2245
Transmit
CAN-ID Type Length Data Cycle Ti... Count Trigger Comment
111h 8 31 31 31 31 31 31 31 31 10 49630 Time
222h 8 32 32 32 32 32 32 32 32 20 24815 Time
333h 8 33 33 33 33 33 33 33 33 30 16544 Time
444h 8 34 34 34 34 34 34 34 34 40 12408 Time
555h 8 35 35 35 35 35 35 35 35 50 9926 Time
666h 8 36 36 36 36 36 36 36 36 60 8272 Time
Connected to hardware PCAN-USB, Device ID FFh
Bit rate: 500 kbit/s
Status: OK

8. 4G 拨号上网测试

默认 SDK 已集成移远 4G 模块驱动，支持移远系列 GSM 模块上网，需要手动拨号下即可上网；

注意：WIFI 模块和 4G 模块共用了同一个 USB，测试 4G 时不要接 WIFI 模块；

指令如下：4G-quectel-CM -s 3gnet &

```
root@TinaLinux:~# 4G-quectel-CM -s 3gnet &
root@TinaLinux:~# [02-23_17:05:45:753] WCDMA&LTE_QConnectManager_Linux&Android_V1.1.34
[02-23_17:05:45:753] 4G-quectel-CM profile[1] = 3gnet///0, pincode = (null)
[02-23_17:05:45:755] Find /sys/bus/usb/devices/1-1.4 idVendor=2c7c idProduct=0125
[02-23_17:05:45:755] Find /sys/bus/usb/devices/1-1.4:1.4/net/usb0
[02-23_17:05:45:755] Find usbnet_adapter = usb0
[02-23_17:05:45:755] Find /sys/bus/usb/devices/1-1.4:1.4/GobiQMI/qcqmi0
[02-23_17:05:45:755] Find qmichannel = /dev/qcqmi0
[02-23_17:05:45:779] Get clientWDS = 7
[02-23_17:05:45:812] Get clientDMS = 8
[02-23_17:05:45:844] Get clientNAS = 9
[02-23_17:05:45:876] Get clientUIM = 10
[02-23_17:05:45:908] Get clientWDA = 11
[02-23_17:05:45:940] requestBaseBandVersion EC20CEHDLGR06A09M1G
[02-23_17:05:46:036] requestGetSIMStatus SIMStatus: SIM_READY
[02-23_17:05:46:036] requestSetProfile[1] 3gnet///0
[02-23_17:05:46:099] requestGetProfile[1] 3gnet///0
[02-23_17:05:46:133] requestRegistrationState2 MCC: 460, MNC: 11, PS: Attached, DataCap: LTE
[02-23_17:05:46:164] requestQueryDataCall IPv4ConnectionStatus: DISCONNECTED
[02-23_17:05:46:228] requestRegistrationState2 MCC: 460, MNC: 11, PS: Attached, DataCap: LTE
[02-23_17:05:46:676] requestSetupDataCall WdsConnectionIPv4Handle: 0x872544e0
[02-23_17:05:46:772] requestQueryDataCall IPv4ConnectionStatus: CONNECTED
[02-23_17:05:46:804] ifconfig usb0 up
[02-23_17:05:46:814] busybox udhcpc -f -n -q -t 5 -i usb0
udhcpc: started, v1.27.2
udhcpc: sending discover
udhcpc: sending select for 10.239.142.146
udhcpc: lease of 10.239.142.146 obtained, lease time 7200
[02-23_17:05:47:031] udhcpc: ifconfig usb0 10.239.142.146 netmask 255.255.255.252 broadcast +
[02-23_17:05:47:036] udhcpc: setting default routers: 10.239.142.145

root@TinaLinux:~# ping www.baidu.com
PING www.baidu.com (183.2.172.185): 56 data bytes
64 bytes from 183.2.172.185: seq=0 ttl=52 time=56.046 ms
64 bytes from 183.2.172.185: seq=1 ttl=52 time=57.442 ms
64 bytes from 183.2.172.185: seq=2 ttl=52 time=48.458 ms
64 bytes from 183.2.172.185: seq=3 ttl=52 time=48.115 ms
64 bytes from 183.2.172.185: seq=4 ttl=52 time=56.016 ms
64 bytes from 183.2.172.185: seq=5 ttl=52 time=48.676 ms
64 bytes from 183.2.172.185: seq=6 ttl=52 time=107.061 ms
64 bytes from 183.2.172.185: seq=7 ttl=52 time=140.095 ms
^C
--- www.baidu.com ping statistics ---
```

拨号上网

ping 百度

9. RS485 测试

RS485 接口驱动节点：/dev/ttyS3

10. 扩展 GPIO

开发板顶部的双排针是引出的 GPIO（PE0~PE9），其中 PE2~PE7 默认作 3 路串口使用，PE0,PE1,PE8,PE9 做普通 GPIO 使用；

① 开发板串口的分配：节点为/dev/ttySN(1..5)

UART1 debug 串口
UART2 扩展 GPIO(PE2,PE3)
UART3 RS485
UART4 扩展 GPIO(PE4,PE5)
UART5 扩展 GPIO(PE6,PE7)

② 普通 GPIO：(PE0,PE1,PE8,PE9)

GPIO 输出使用方法：

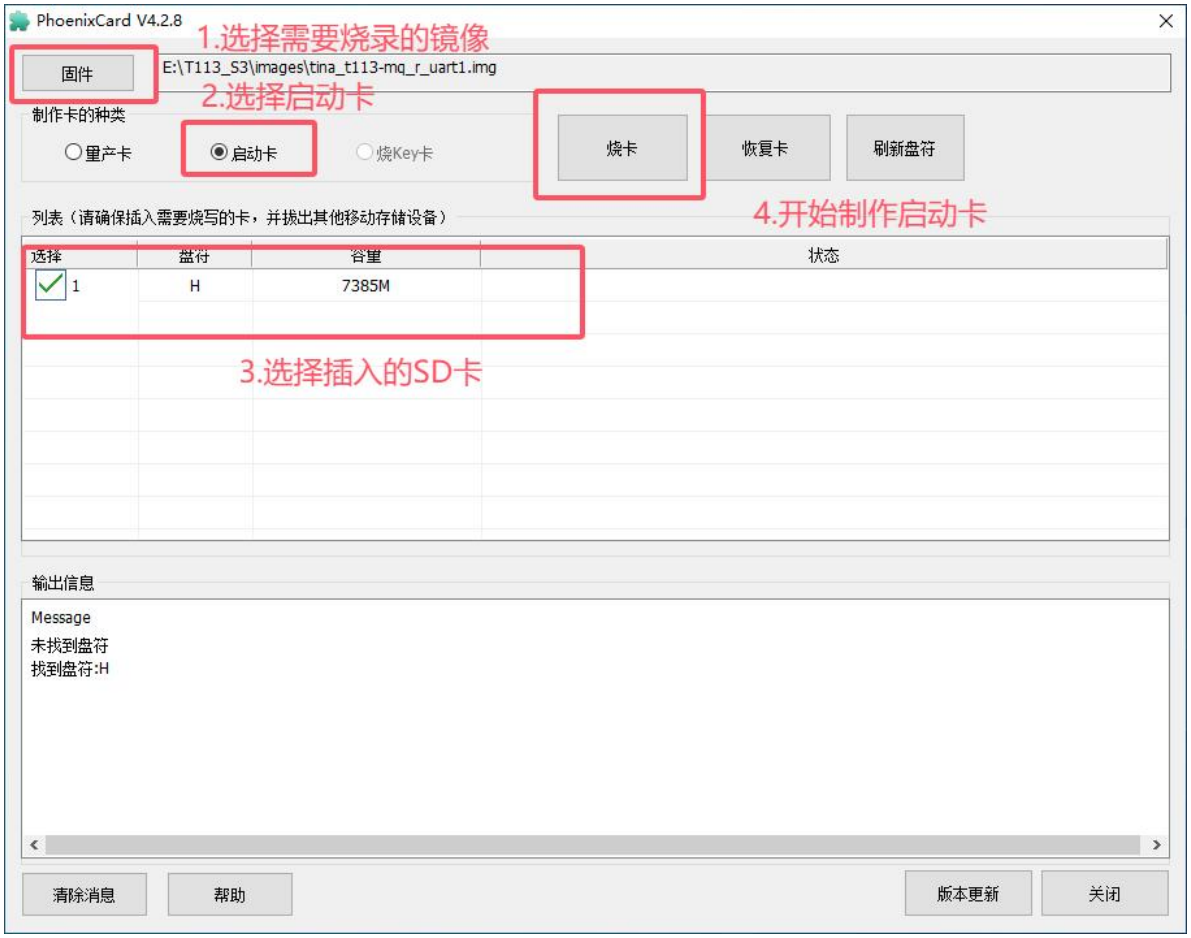
```
##
# GPIO_PE0 输出高电平
echo 1 > /sys/class/leds/GPIO_PE0/brightness
# GPIO_PE0 输出低电平
echo 0 > /sys/class/leds/GPIO_PE0/brightness

# GPIO_PE1 输出高电平
echo 1 > /sys/class/leds/GPIO_PE1/brightness
# GPIO_PE1 输出低电平
echo 0 > /sys/class/leds/GPIO_PE1/brightness
```

```
# GPIO_PE8 输出高电平
echo 1 > /sys/class/leds/GPIO_PE8/brightness
# GPIO_PE8 输出低电平
echo 0 > /sys/class/leds/GPIO_PE8/brightness

# GPIO_PE9 输出高电平
echo 1 > /sys/class/leds/GPIO_PE9/brightness
# GPIO_PE9 输出低电平
echo 0 > /sys/class/leds/GPIO_PE9/brightness
```

三．烧录到 Nand flash

1. 安装 usb 驱动：Tools/全志 USB 烧录驱动 20201229.zip
2. 解压 AllwinnertechPhoeniSuit.zip，打开线刷工具 PhoenixSuit.exe，提前将 usb 插入到 USB-HOST 口

3. 按住 FEL 按键，然后按下 RESET 按键松开，出现开始烧录松开 FEL 按键。

4. 烧录完成后会自动重启，观察 Debug 串口日志；

四．烧录到 TF card 启动（如果 Nand 储存不够时，可以用 sd 卡启动）